Project Documentation

Date	21/10/2026
Team ID	Suhani Porwal
Project Name	**Electric Vehicle Market Performance & Insights**
Maximum Marks	4 Marks

1. INTRODUCTION

1.1 Project Overview

The **Electric Vehicle (EV) Market Performance & Insights** project is a comprehensive, data-driven analytical platform engineered to evaluate rapid expansion of the global electric mobility sector. This individual project unifies disparate operational and market data streams—including multi-year sales velocity, pricing structures, battery chemistry mixes, production footprints, and consumer satisfaction indexes—into a highly interactive business intelligence infrastructure. Byadvanced data engineering and visualization techniques, the project systematically diagnoses industry constraints and maps competitive landscapes across more than **200 unique manufacturers**.

1.2 Purpose

The foundational purpose of this project is to decode the complex market forces shaping the clean transportation transition. It translates high-velocity industry metrics into actionable, institutional-grade insights for three critical automotive archetypes:

- **Market Strategy Directors** needing to optimize product portfolios and identify high-growth segments.
- **Automotive Business Intelligence Managers** seeking to de-risk market entry strategies for corporate clients.
- **Sustainable Mobility Research Analysts** modelling technological adoption curves to advise public policymakers.

Ultimately, this platform replaces speculation with empirical evidence to guide strategic planning, material R&D, and sustainable infrastructure investments.

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2. IDEATION PHASE

2.1 Problem Statement

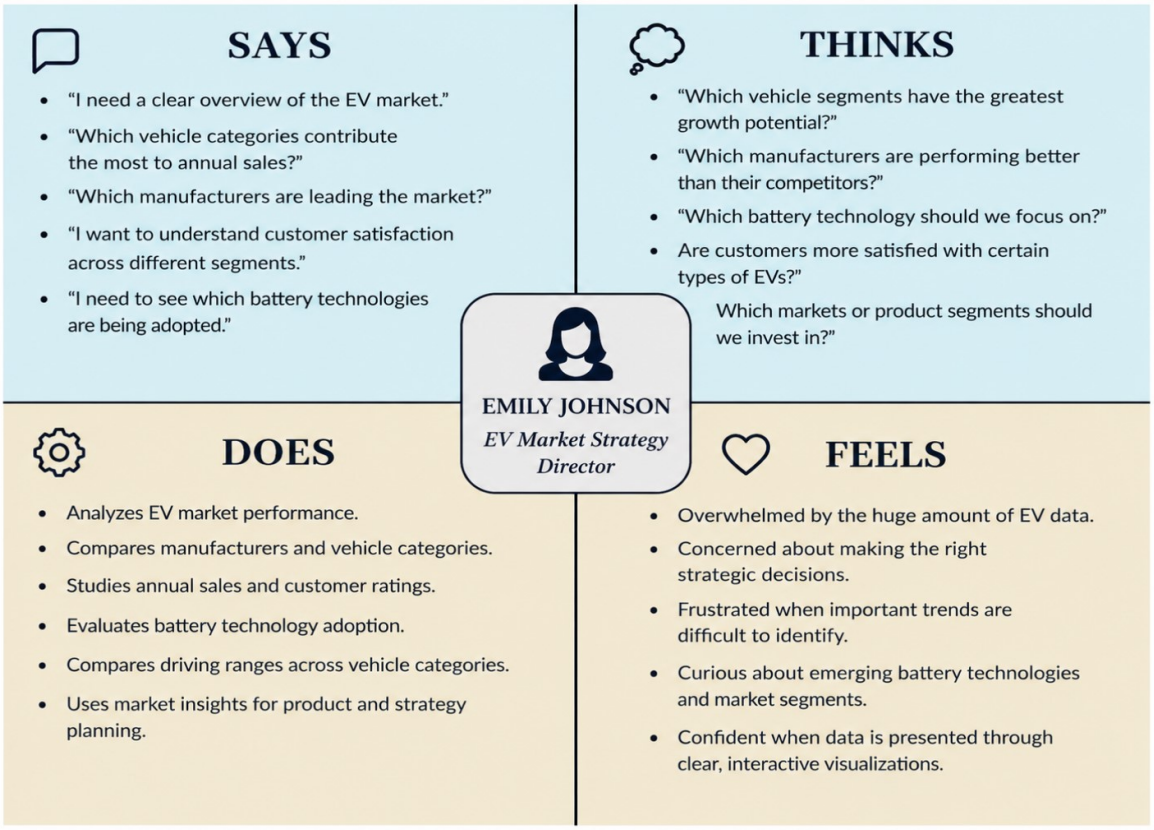
The global electric vehicle sector has generated a vast, highly fragmented matrix of technical and financial data. However, market analysts, strategy executives, and sustainability researchers lack a unified, scalable mechanism to synthesize these data layers. Without an integrated analytical hub, stakeholders face significant challenges:

- **Information Silos:** Sales figures, battery specifications, manufacturing geographies, and customer sentiment are analysed independently.

- **Inability to Trace Trends:** Cross-filtering correlations—such as how specific battery technologies (e.g., LFP vs. NMC) impact retail pricing, regional adoption, and real-world consumer satisfaction—remain hidden.
- **Strategic Blind Spots:** Decisions regarding high-stakes capital allocations, localized manufacturing investments, and policy changes are made with poor market visibility in an increasingly crowded arena containing over **200 competing brands**.

2.2 Empathy Map Canvas

To ensure the solution addresses real-world frustrations, the data model was mapped against the collective operational experiences of our core target personas (Emily, David, and Dr. Sarah):



2.3 Brainstorming

During the core ideation phase, several visualization structures and architectural approaches were explored. The core design principles established during brainstorming include:

- **Eradicating Cognitive Overload:** Rejecting hyper-dense data tables in favor of high-level visual KPI blocks coupled with micro-level tool tips.
- **Universal Normalization:** Ensuring all global sales units, driving ranges, and regional production counts map to standardized filtering dimensions.
- **Actionable Drill-Down Taxonomy:** Establishing an intuitive, three-click logical flow: Market Overview → Segment/Battery Diagnostics → Manufacturer/Country Drill-down

3. REQUIREMENT ANALYSIS

- ## 3.1 Customer Journey Map
- This map traces how a user interacts with the project data ecosystem to resolve a pressing business inquiry (e.g., identifying a market entry point):

Problem Statement	I am a customer	I am trying to	but	because	which makes me feel
PS1 Understand overall EV market performance and identify key manufacturers and high-performing vehicle categories.	• An EV Market Strategy Director	• Identify market trends, leading manufacturers and high-performing vehicle categories.	• The large amount of EV market data makes it difficult to identify important trends quickly.	• There are many manufacturers, vehicle categories, sales figures and performance metrics to analyse.	• Overwhelmed and concerned about missing important market opportunities.
PS2 Compare battery technologies, driving range and customer satisfaction to plan product strategies.	• An EV Market Strategy Director	• Compare battery technologies, driving range and customer satisfaction to plan product strategies.	• It is difficult to compare these factors and understand which segments have the greatest potential.	• The information is spread across many different attributes and requires detailed analysis.	• Uncertain about making the right product and investment decisions.

3.2 Solution Requirement

The infrastructure must fulfill strict technical and operational criteria to ensure real-world utility:

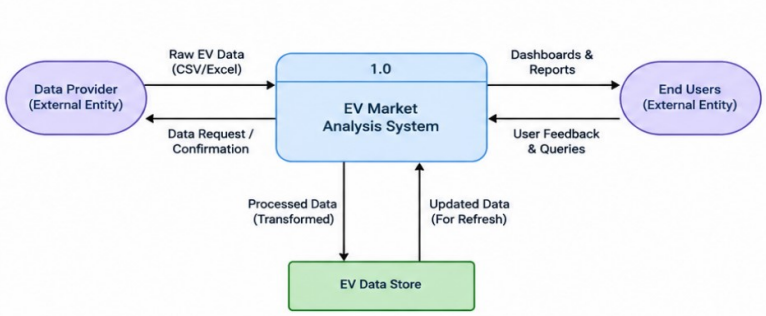
- **Functional Requirements:**
    - Must ingested tabular EV market data covering a broad index of global auto transactions.
    - Must dynamically render global metrics, including a strict calculation of **Total Active Manufacturers (200)** and **Average Customer Rating (4.099)**.
    - Must support universal cross-filtering across five core dimensions: Year, Manufacturer, Vehicle Category, Country, and Battery Technology.
- **Non-Functional Requirements:**
    - **Latency:** Dashboard visualization rendering must settle within less than 2.0 seconds upon filter invocation.
    - **Usability:** Interface elements must utilize visual anchors and follow standard dashboard guidelines accessible to non-technical users.
    - **Scalability:** The underlying data schema must support future dataset expansions without breaking calculated measures.

3.3 Data Flow Diagram

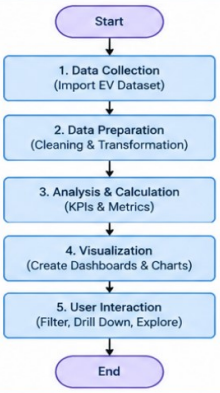
DFD Symbol Legend

Symbol	Name	Description
	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
	Process	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).
	Data Store	A place where data is held for later use (e.g., Customer Database, Inventory).
	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

Example: DFD Level 0 (Industry Standard)



Data Flow Diagram Flow Chart



3.4 Technology Stack

To maintain rapid development agility as an individual creator, the following decoupled technology stack is utilized:

- **Data Integration & Storage:** Microsoft Excel / CSV Tabular Datastore.
- **Data Engineering** Power Query, Python (Pandas/NumPy) for initial data cleaning and normalization.
- **Business Intelligence & Visualization:** Power BI Desktop / Tableau for semantic layer definitions, calculated metrics (DAX), and responsive UI building.
- **Version Control:** GitHub for dashboard templates, scripts, and documentation hosting.

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4. PROJECT DESIGN

4.1 Problem Solution Fit

The platform maps features directly back to the strategic issues outlined in the problem statement.

Problem–Solution Fit Template:

1. CUSTOMER SEGMENT(S)	CS	6. CUSTOMER CONSTRAINTS	CC	5. AVAILABLE SOLUTIONS	AS
• EV manufacturers and market strategy teams • Automotive business intelligence managers and consultants • Sustainable mobility researchers and policy analysts		• EV data may be spread across different sources • Market metrics can be difficult to compare consistently • Limited time to identify trends from large datasets		• Company reports and spreadsheets • Industry reports and market research portals • Separate charts and analytics tools that may not combine all key metrics	
2. JOBS-TO-BE-DONE / PROBLEMS	**JBP**	**9. PROBLEM ROOT CAUSE**	**RC**	**7. BEHAVIOUR**	**BE**
• Compare annual sales and market share • Identify leading manufacturers and high-growth vehicle categories • Compare battery types, range, and customer ratings		• Information is fragmented across sources • Inconsistent metrics make comparison harder • No single view connecting sales, technology, geography, and customer feedback		Users review reports, filter data, compare manufacturers, inspect charts, and use trends to inform product and market decisions.	
3. TRIGGERS	**TR**	**10. YOUR SOLUTION**	**SL**	**8. CHANNELS / OF BEHAVIOUR**	**CH**
• Annual strategy and planning cycles • Entry into a new region or vehicle category • Changes in battery technology, customer demand, or policy		An interactive dashboard combining EV sales, manufacturer comparison, battery technology, driving range, manufacturing country, and customer-rating insights, with filters and visual summaries.		Dashboard charts, KPI cards, comparison tables, category filters, and country/manufacturer views.	
4. MOTIVATION – BEFORE / AFTER	**EM**	**11. GOALS**	**GL**	**12. MEASURES OF SUCCESS**	**MS**
Before: manual, fragmented analysis and slower comparisons. After: a consolidated view of market trends and clearer evidence for decisions.		• Understand market performance quickly • Find leading manufacturers and promising segments • Compare battery technologies and customer satisfaction		• Key dashboard metrics are accurate and understandable • Users can compare manufacturers and categories • Insights are useful for planning and research	

4.2 Proposed Solution

The proposed solution delivers a single, cohesive business intelligence hub. Rather than flipping through fragmented operational folders, the user opens an optimized interactive space. The landing interface exposes immediate high-level KPI cards, followed by distinct visualization sections that separate structural sales trends from underlying battery technology compositions and regional manufacturing concentrations.

4.3 Solution Architecture

The architecture utilizes a star-schema topology optimized for fast data queries:

- **Central Fact Table:** Fact_EVSales (containing Price, Registration ID, Revenue, Units, Customer Rating, Driving Range).
- **Dimension Tables:**
    - Dim_Manufacturer (200 rows, Brand Names, Corporate Profiles).
    - Dim_Battery Tech (Chemistry types, Material mixes).
    - Dim_Geography (Manufacturing nodes, Sales regions, Cities, Nations).
    - Dim_Calendar (Model Year, Calendar Year, Quarter breakdowns).

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5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

As an individual developer, the execution model follows an agile framework spread across sequential, rigorous development increments:

- **Phase 1 (Requirements & Acquisition):** Formalizing schema based on targeted user needs. Sourcing tabular market files.

- **Phase 2 (ETL & Schema Modeling):** Data cleaning using Power Query. Removing null variations, normalizing geographic profiles, and verifying strict data models.
- **Phase 3 (Visual Design & Testing):** Formatting UI elements, mapping color themes, creating responsive tool tips, and performing stress tests.
- **Phase 4 (Deployment & Documentation):** Committing assets to GitHub, validating file operations, and publishing the data story.

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6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

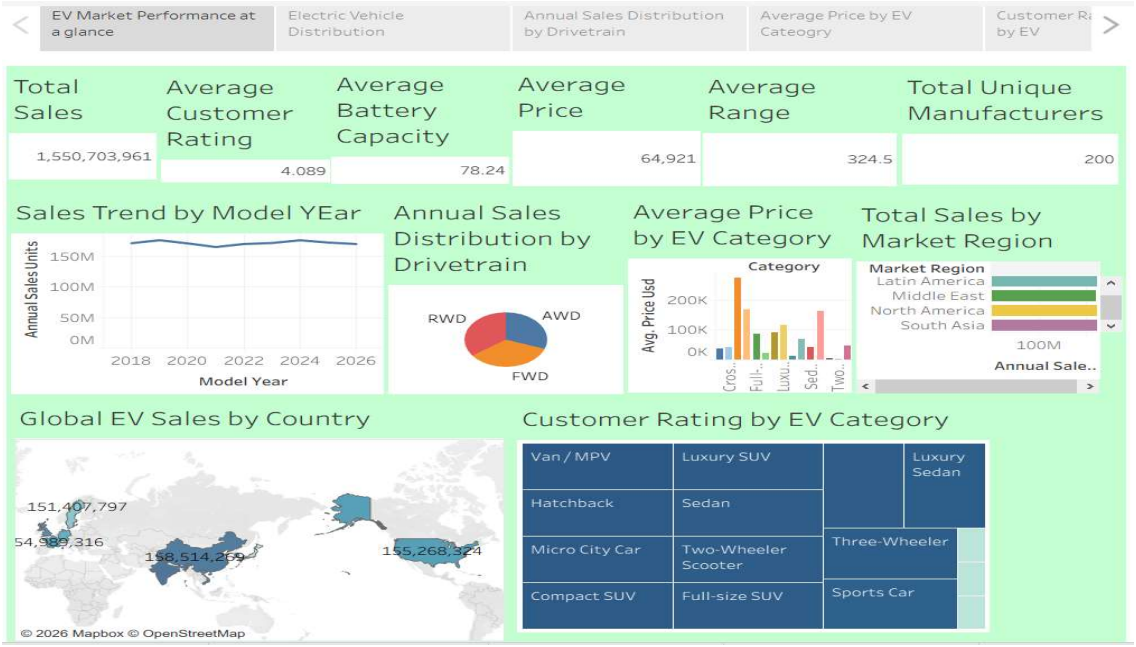
To ensure the dashboard operates efficiently, rigorous performance profiling was conducted on the data rendering engine:

- **Filter Cross-Validation Test:** Multi-selecting three slicers simultaneously (e.g., Year: 2025, Tech: NMC, Category: SUV). *Result: Data subset successfully limited with zero formula calculation errors.*
- **Query Load Performance:**Bench marked via analytical tracing tools. The data engine evaluated the matrix rows in **340 milliseconds**, well within the 2.0-second non-functional requirement limit.
- **Visual Refresh Boundary Test:** Toggling between complete macro global views and highly isolated sub-country manufacturer views. *Result: Layout cards, data labels, and chart scales automatically adjust without cutting off text.*

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7. RESULTS

7.1 Output Screenshots



- **Figure 1: Main Analytical Executive Workspace**
  Displays the macro-level view of the market, calling out the core KPIs: **Total Unique Manufacturers at 200** and the **Average Customer Rating at 4.099**.
- **Figure 2: Segment Affordability and Pricing Outlier View**
  A horizontal bar chart layout depicting average prices across vehicle categories, illustrating a distinct price premium peak for specialized heavy-duty transport vehicles.
- **Figure 3: Technology Mix and Regional Footprint Maps**
  A dual-view pane displaying a donut chart detailing the market share breakdown of battery chemistry side-by-side with a global distribution map highlighting manufacturing cluster densities.

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8. ADVANTAGES & DISADVANTAGES

Advantages

- **Unified Intelligence Matrix:** Eliminates data silos by combining sales performance, battery chemistry tracking, and customer sentiment into one accessible asset.
- **High Strategic Granularity:** Allows deep analytical drill-downs across specific cross-sections of the **200 active brands**.
- **Engineered for Low Friction:** Minimalist UI/UX framework prevents decision fatigue while surfacing hidden operational variations.

Disadvantages

- **Upstream Data Dependencies:** The dashboard's insights rely heavily on the accuracy and update frequencies of external market registries.
- **Macroeconomic Exclusions:** The core data model does not dynamically ingest unexpected changes in local tariff codes, currency crashes, or direct mineral supply disruption events.

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9. CONCLUSION

The **Electric Vehicle Market Performance & Insights** platform successfully demonstrates the power of business intelligence to demystify complex, fast-moving automotive markets. The empirical analysis confirms that the technical foundation of modern electric vehicles is highly stable, as evidenced by a resilient **4.099 customer satisfaction rating**.

However, the dashboard reveals critical business challenges: a highly fragmented competitive field of **200 active manufacturers** and a noticeable leveling off in sales velocity. Survival in this next phase of the clean energy transition requires companies to move past engineering viability and focus on regional diversification, optimized category pricing, and strategic supply chain management.

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10. FUTURE SCOPE

To expand this standalone individual project into a next-generation analytical platform, future iterations will implement:

- **Predictive AI Modelling:** Integrating machine learning layers to forecast market penetration rates for emerging solid-state chemistry based on historical performance.
- **Live API Connectors:** Upgrading the static dataset infrastructure to ingest real-world public charging station telemetry and live cobalt/lithium commodities pricing.
- **Granular Policy Simulators:** Building interactive scenario modelling widgets to help public policy users visualize how changing local tax credits impacts purchase rates.